AI for Humanity, Lecture 7

Fairness

Stephanie Brandl

Agenda

- Binary classification
- Confusion Matrix
- Precision/Recall and type I/II error
- Bias and Fairness
- Different Fairness Metrics
- The impossibility theorem
- Revisiting the impossibility theorem in practice

Resources

Most important papers for today:

- Chouldechova, 2017. "Fair Prediction with Disparate Impact: A Study of Bias in Recidivism Prediction Instruments"
- Buolamwini & Gebru, 2018. "Gender Shades: Intersectional Accuracy Disparities in Commercial Gender Classification"
- Bell et al., 2023. "The Possibility of Fairness: Revisiting the Impossibility Theorem in Practice"

More literature on the website.

Binary classification

and how to evaluate it

Binary classification: a classification problem with binary labels

Examples:

- Loan lending (yes/no)
- Medical testing (sick/not sick)
- Bailout (leave jail/stay in jail)
- Student admission (accept/reject)
- Face recognition (face/no face or male/female)

Binary classification
and how to evaluate it

Classical evaluation metrics:

$$\text{Accuracy: } \frac{\#\text{correctly classified}}{\#\text{all data points}}$$

➡️ does not give us any information what type of errors occur

- positive data points labeled as negative or vice versa?

Better: F1 score! Harmonic mean of precision and recall

$$\text{F1 score} = 2\frac{\text{Precision} \cdot \text{Recall}}{\text{Precision} + \text{Recall}}$$

➡️ wait, what's precision and recall? 🙀

The confusion matrix

		Predicted Labels	
		$\hat{P}$	$\hat{N}$
True Labels	P	True Pos.	False Neg.
	N	False Pos.	True Neg.

Let's consider the example of medical testing. We are testing a group of people for a disease, if the test is positive, this means they very likely have the disease.

- True Positive: test is positive and person is actually sick
- True Negative: test is negative and person is healthy
- False Positive: test is positive but person is healthy
- False Negative: test is negative but person is sick

The confusion matrix

		Predicted Labels	
		$\hat{P}$	$\hat{N}$
True Labels	P	True Pos. (Recall)	False Neg. (Type II err.)
	N	False Pos. (Type I err.)	True Neg.

- TP: test is positive & person is sick

$$\text{Recall/TP rate} = \frac{TP}{P} = \frac{TP}{TP + FN}$$

how many sick people where diagnosed correctly?

- TN: test is negative & person is healthy

$$\text{TN rate} = \frac{TN}{N} = \frac{TN}{TN + FP}$$

how many healthy people where diagnosed correctly?

- FP: test is positive but person is healthy

$$\text{Type I err./FP rate} = \frac{FP}{N} = \frac{FP}{TN + FP}$$

how many healthy people were misclassified as sick?

- FN: test is negative but person is sick

$$\text{Type II err./FN rate} = \frac{FN}{P} = \frac{FN}{TP + FN}$$

how many sick people were diagnosed as healthy?

The confusion matrix

		Predicted Labels	
		$\hat{P}$	$\hat{N}$
True Labels	P	True Pos. (Recall)	False Neg. (Type II err.)
	N	False Pos. (Type I err.)	True Neg.
Prevalence		Pos. Pred. Value (Precision)	

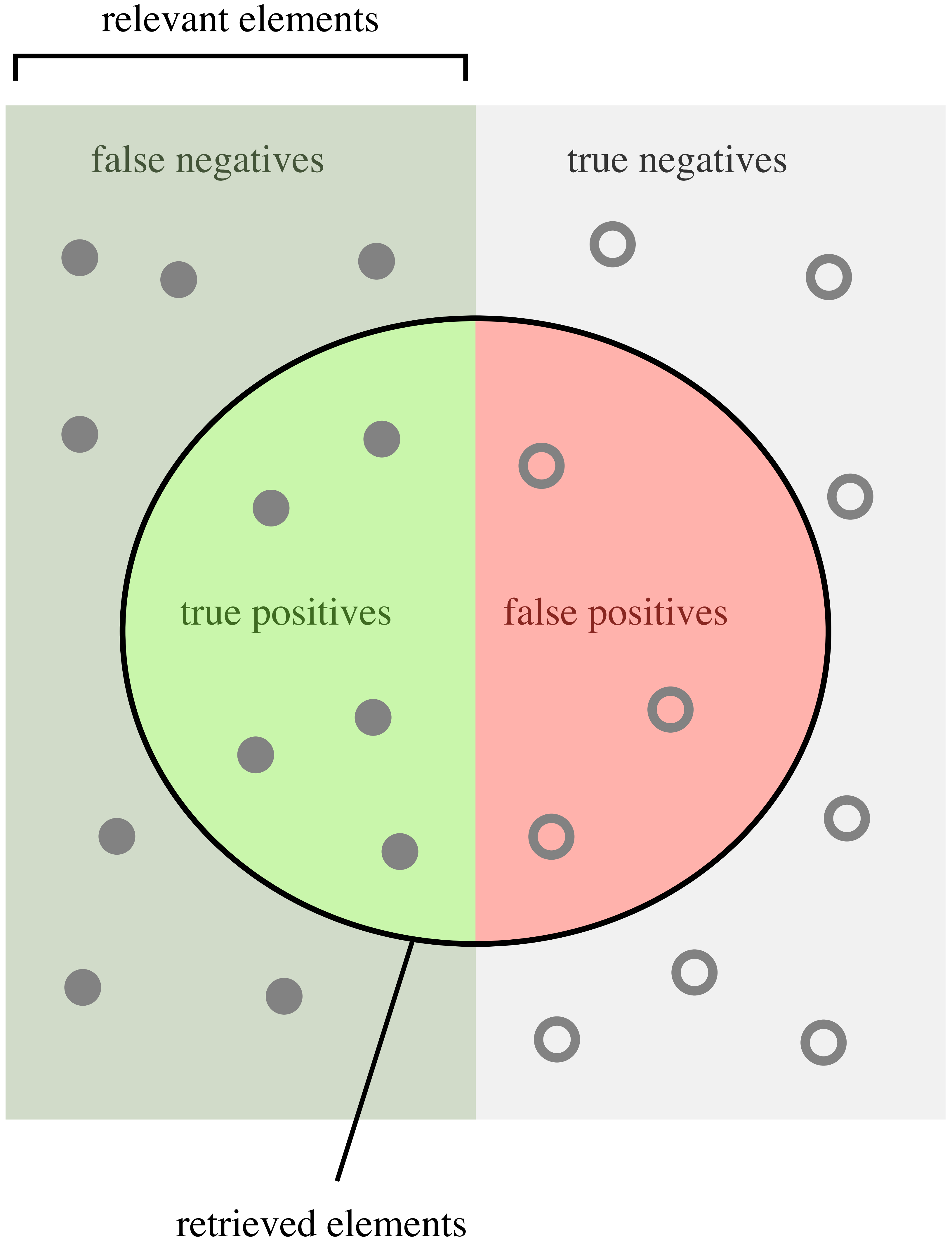
$$\text{Prevalence} = \frac{P}{P+N}$$

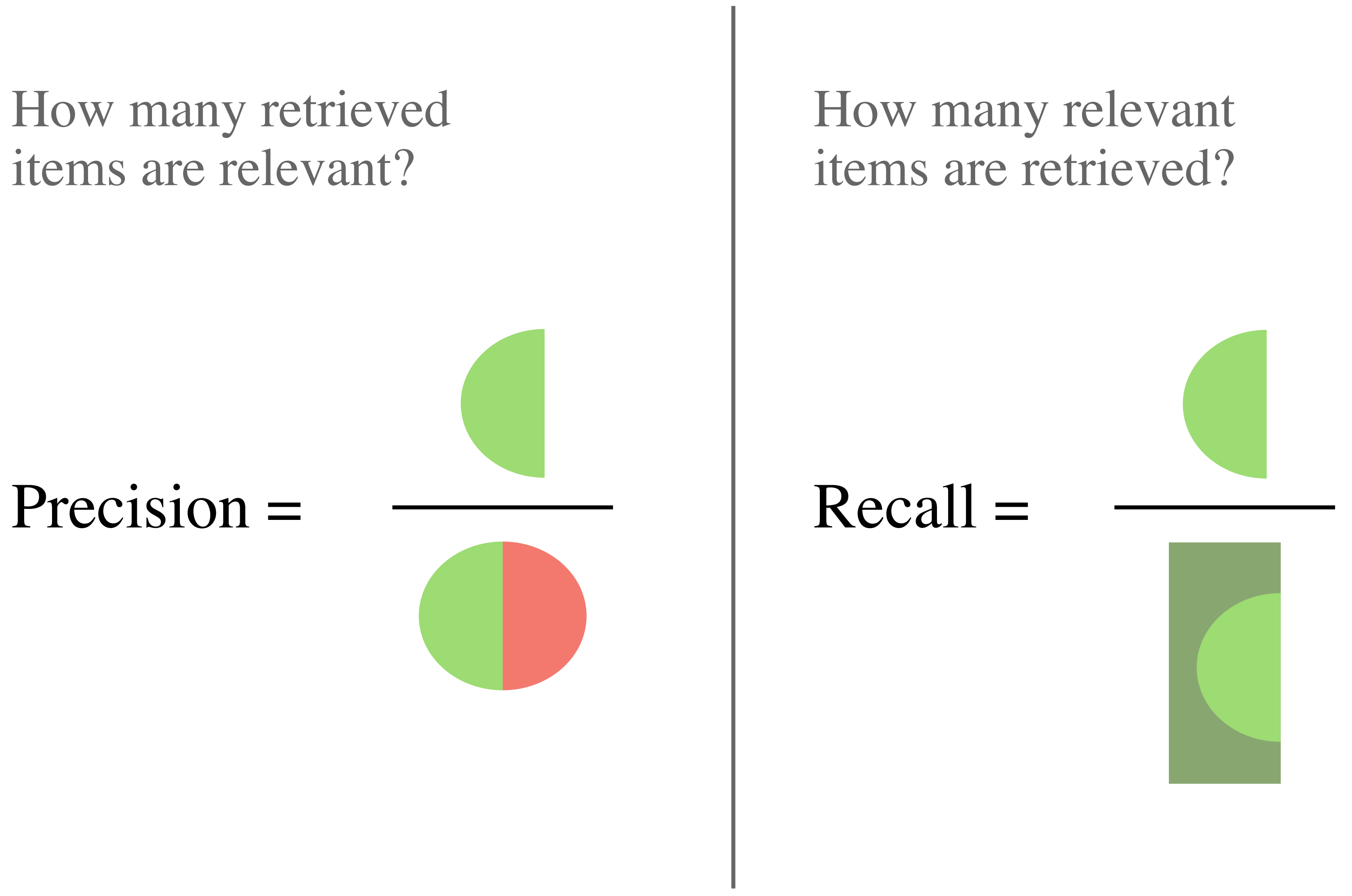
How often does the disease occur among the tested population?

$$\text{Precision} = \frac{TP}{TP+FP}$$

How many of the positive tested are actually sick?

Precision/Recall





How many of the positive tested are actually sick?

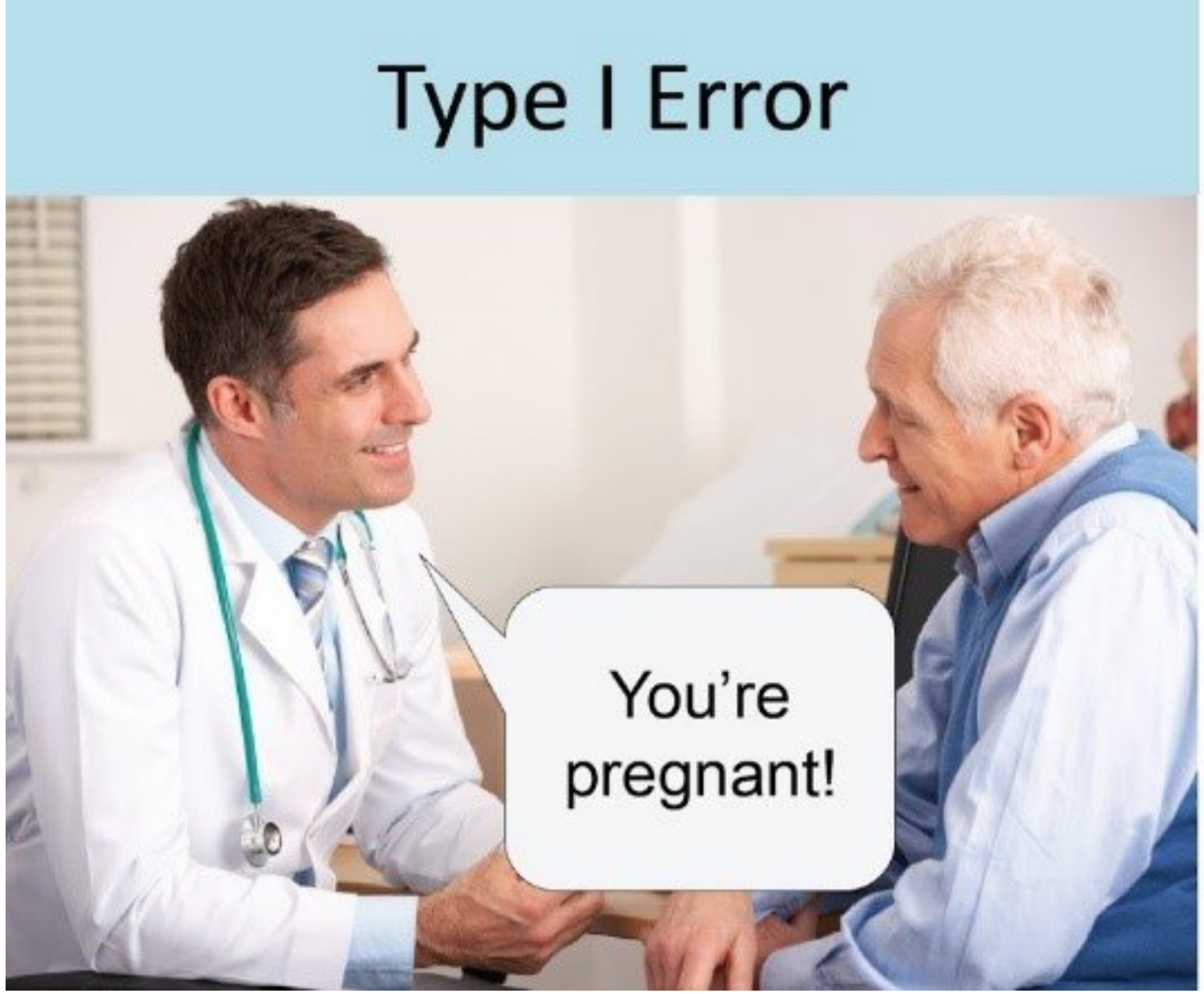
+→🤧

How many sick people where diagnosed correctly?

🤧→+

$$\text{F1 score} = 2\frac{\text{Precision} \cdot \text{Recall}}{\text{Precision} + \text{Recall}}$$

Type I/II error



False Positive

False Negative

Which error do we try to minimize?

- Depending on the classification problem and the perspective, we want to minimize different errors
  - Type I error: False Positive vs.
  - Type II error: False Negative
- Let's revisit our examples and decide:
  - (Bank) Loan lending (yes/no)
  - (Patient) Medical testing (sick/not sick)
  - (State) Bailout (leave jail/stay in jail)
  - (University) Student admission (accept/reject)
  - (Cat-lover) Image recognition (cat/no cat)

And how does all of this relate to fairness?

We will get there!

Bias vs. (Un)Fairness

- Bias and Fairness are often conflated in the literature and used interchangeably
- We will use different definitions:
  - Bias in a Language Model:
    When language is encoded systematically different in a model
    e.g.: the word "vector computer" programmer is closer to the word vector "man" than "woman"
  - (Un)Fairness in a Language Model:
    When a model performs better in a specific task for one group than the other
    e.g.: facial recognition performing better for light-skinned people than dark-skinned people
- Bias can lead to unfair outcomes but they have also found to be unrelated 🤯

Representational Bias
Sampling Bias

Sampling Bias: occurs when the model learns from biased sampling from the population

- face/skin recognition: some skin tones are more present than others in the training data
  - might lead to skewed representation of faces
- credit lending: information of repayment is only known for people that were granted a loan
  - very difficult to make any assumptions about people who were not granted a loan (false negatives/true negatives)

Representational Bias
Historical Bias

Historical Bias: “the already existing bias and socio-technical issues in the world” [Mehrabi et al., 2022]

- image search “CEO United States” shows predominantly white males
- credit lending: based on past biased decisions by humans
- gender bias: existing income disparities

Whether we want to mirror these disparities in our model depends on the context and the application

Three definitions of Fairness

There is not a single definition of fairness!

- this depends on the task, data and the goal
- different demographic groups might require different fairness definitions
- we will revisit three of them here

⚠️ Disclaimer ⚠️

Similar to the confusion matrix, there is a lot of terminology that sounds different but means the same!

Three definitions of Fairness

1. Group Fairness

Decisions $\hat{Y}$ should be independent of any sensitive attribute A, where $\mathscr{A}$ is the set of sensitive attributes.

$$\hat{Y} \perp\!\!\!\perp A$$

$$P(\hat{Y} = 1 \,|\, A = a) = P(\hat{Y} = 1 \,|\, A = b) \qquad \forall a, b \in \mathscr{A}$$

Examples:

- loans: whether someone receives a loan should be independent of their gender
- face recognition: a person's face should be recognized by a classifier independent of their skin color

Metric:

- Group disparity: difference in F1 scores between groups

Limitations:

- if the prevalence is different across groups, a perfect classifier cannot achieve group fairness
  - for instance, if women tend to pay back their loan more reliably, a classifier optimising for accuracy would favour women over men

Group Fairness

Soap dispenser

Three definitions of Fairness

2. Equalized Odds

(Positive) Decisions $\hat{Y}$ should be independent of the protected attributes for both classes

$$\hat{Y} \perp\!\!\!\perp A \mid Y$$

$$P(\hat{Y} = 1 \mid A = a, Y = y) = P(\hat{Y} = 1 \mid A = b, Y = y)$$

→ same true positive and false positive rate across groups

- **Equality of Opportunity**: same TPR/Recall, **Predictive Equality**: same FPR

Example of loan lending ($A =$ gender):

- the percentage of men who are given loans that they will pay back must also match the corresponding percentage of women (same TPR)
- similarly the percentage of men who are wrongly denied loans even though they would have paid it back must match the corresponding percentage for women (same FPR)
- this does not require that the same percentage of male/female applicants will receive loans, e.g., if women pay back loans with higher frequency than men, the predictor would be allowed to deny loans to men more often than women

Use case: Facial Recognition

Gender Shades: Intersectional Accuracy Disparities in Commercial Gender Classification*

Joy Buolamwini JOYAB@MIT.EDU
MIT Media Lab 75 Amherst St. Cambridge, MA 02139

Timnit Gebru TIMNIT.GEBRU@MICROSOFT.COM
Microsoft Research 641 Avenue of the Americas, New York, NY 10011

Buolamwini & Gebru, PMLR 2018

Use case: Facial Recognition

- Task: Face Recognition
  classify if an image contains a female or a male face
  → usually only binary gender is considered, task itself potentially problematic
- Problem setting:
  Many datasets are imbalanced with respect to gender and skin types
  → representational bias
  This leads to performance discrepancies
  → “unfairness”
- Paper:
  Collect a more balanced dataset from 3 European and 3 African countries
  and test commercial gender classifiers on their new dataset

Three definitions of Fairness

2. Equality of Opportunity

True Positive Rate/ Recall

Female

Male

Different gender classifiers

Classifier	Metric	All	F	M	Darker	Lighter	DF	DM	LF	LM
MSFT	TPR(%)	93.7	89.3	97.4	87.1	99.3	79.2	94.0	98.3	**100**
Face++	TPR(%)	90.0	78.7	99.3	83.5	95.3	65.5	**99.3**	90.2	99.2
IBM	TPR(%)	87.9	79.7	94.4	77.6	96.8	65.3	88.0	92.9	**99.7**

Buolamwini & Gebru, PMLR 2018

Three definitions of Fairness

2. Equality of Opportunity

Recall Female Male

Different gender classifiers

Classifier	Metric	All	F	M	Darker	Lighter	DF	DM	LF	LM
MSFT	TPR(%)	93.7	89.3	97.4	87.1	99.3	79.2	94.0	98.3	**100**
Face++	TPR(%)	90.0	78.7	99.3	83.5	95.3	65.5	**99.3**	90.2	99.2
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Buolamwini & Gebru, PMLR 2018

Three definitions of Fairness

2. Equality of Opportunity

Recall

Female Male

Different gender classifiers

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IBM	TPR(%)	87.9	79.7	94.4	77.6	96.8	65.3	88.0	92.9	**99.7**

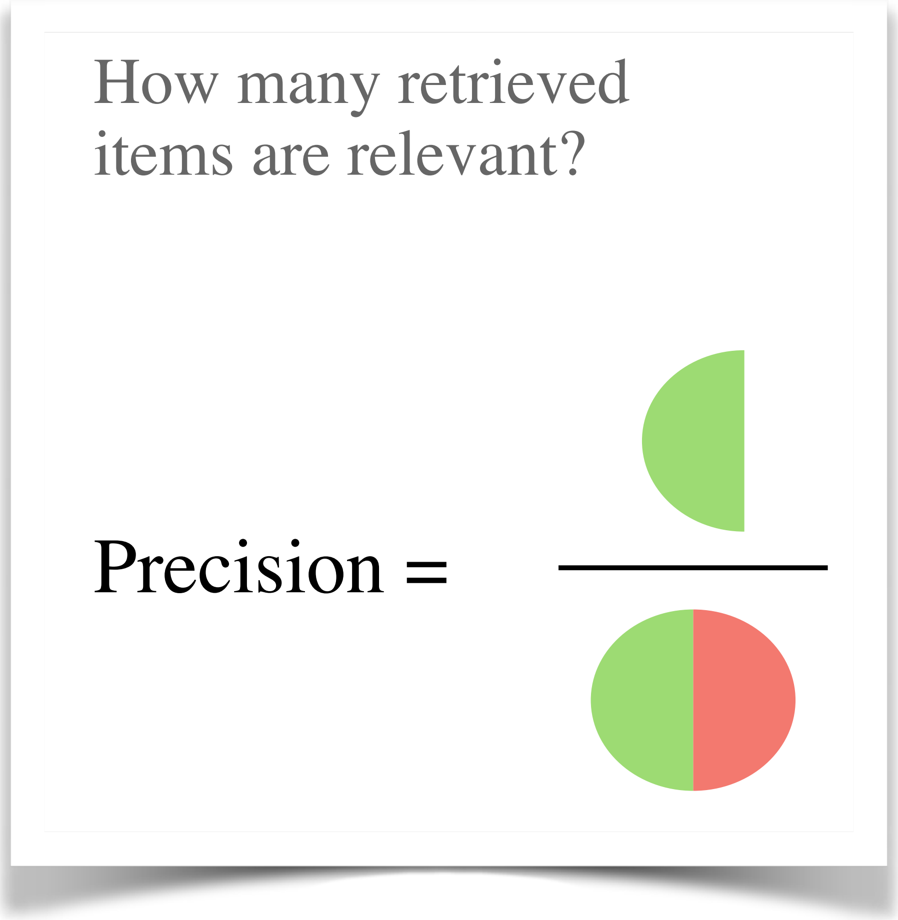
- higher recall for male vs. female lighter vs. darker-skin
- intersectional analysis shows lowest recall for darker females (highest for lighter males)
- darker-skinned women are more often misclassified as men than any other group

Buolamwini & Gebru, PMLR 2018

Three definitions of Fairness

3. Predictive Parity

Predicted positive outcomes have the same precision across attributes.



$$P(Y = 1 \,|\, A = a, \hat{Y} = 1) = P(Y = 1 \,|\, A = b, \hat{Y} = 1)$$

Example of student admission:

Among the students who were accepted, the same ratio will graduate for both groups.

Three definitions of Fairness

3. Predictive Parity

True Positive Rate/Recall
Positive Pred. Val./Precision

Female
Male

Different gender classifiers

Classifier	Metric	All	F	M	Darker	Lighter	DF	DM	LF	LM
MSFT	TPR(%)	93.7	89.3	97.4	87.1	99.3	79.2	94.0	98.3	**100**
	PPV (%)	93.7	96.5	91.7	87.1	99.3	92.1	83.7	**100**	98.7
Face++	TPR(%)	90.0	78.7	99.3	83.5	95.3	65.5	**99.3**	90.2	99.2
	PPV (%)	90.0	98.9	85.1	83.5	95.3	98.8	76.6	**98.9**	92.9
IBM	TPR(%)	87.9	79.7	94.4	77.6	96.8	65.3	88.0	92.9	**99.7**
	PPV (%)	87.9	92.1	85.2	77.6	96.8	82.3	74.8	**99.6**	94.8

Buolamwini & Gebru, PMLR 2018

Three definitions of Fairness

3. Predictive Parity

True Positive Rate/Recall

Positive Pred. Val./Precision

Female

Male

Different gender classifiers

Classifier	Metric	All	F	M	Darker	Lighter	DF	DM	LF	LM
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Buolamwini & Gebru, PMLR 2018

Three definitions of Fairness

3. Predictive Parity

Recall

Precision

Female Male

Different gender classifiers

Classifier	Metric	All	F	M	Darker	Lighter	DF	DM	LF	LM
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	PPV (%)	87.9	92.1	85.2	77.6	96.8	82.3	74.8	**99.6**	94.8

- higher precision for female vs. male lighter vs. darker-skin
- intersectional analysis shows lowest precision for darker males (highest for lighter females)

- whenever the model classifies a (light-skinned) woman, it is more often right/certain compared to classifying a (dark-skinned) man

Buolamwini & Gebru, PMLR 2018

The Impossibility Theorem

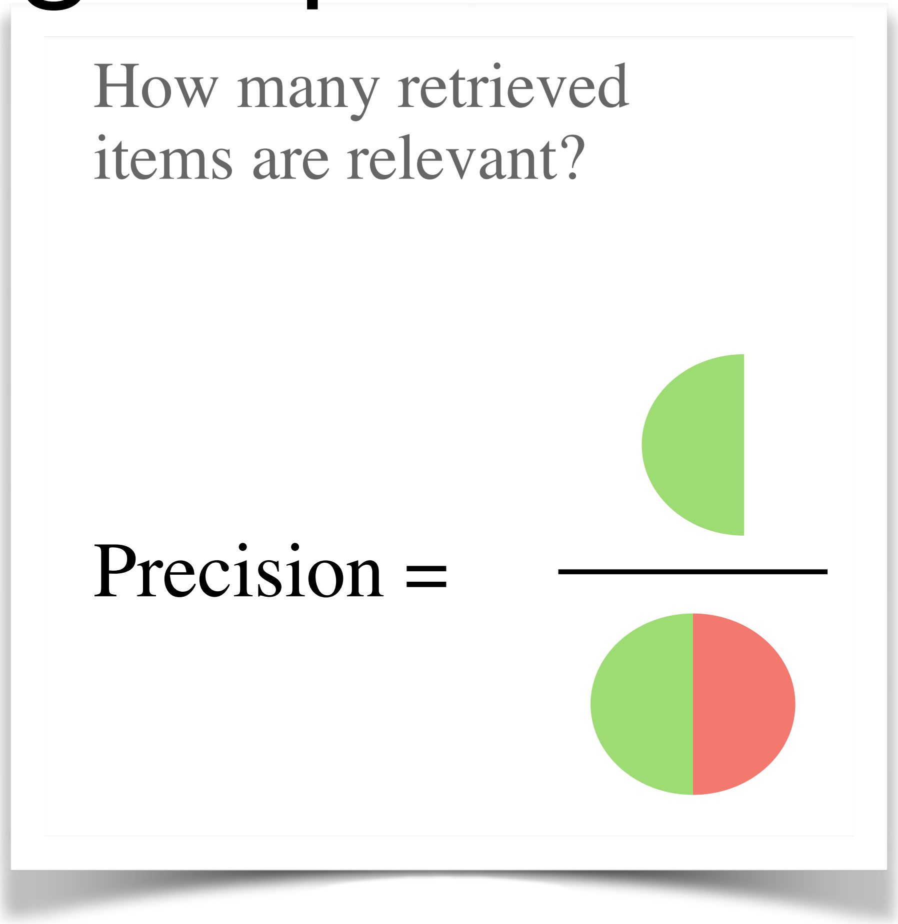
The Impossibility Theorem

The impossibility theorem states that we cannot achieve equalized odds and predictive parity at the same time!

Equalized odds: same true positive and false positive rate across groups

⚡

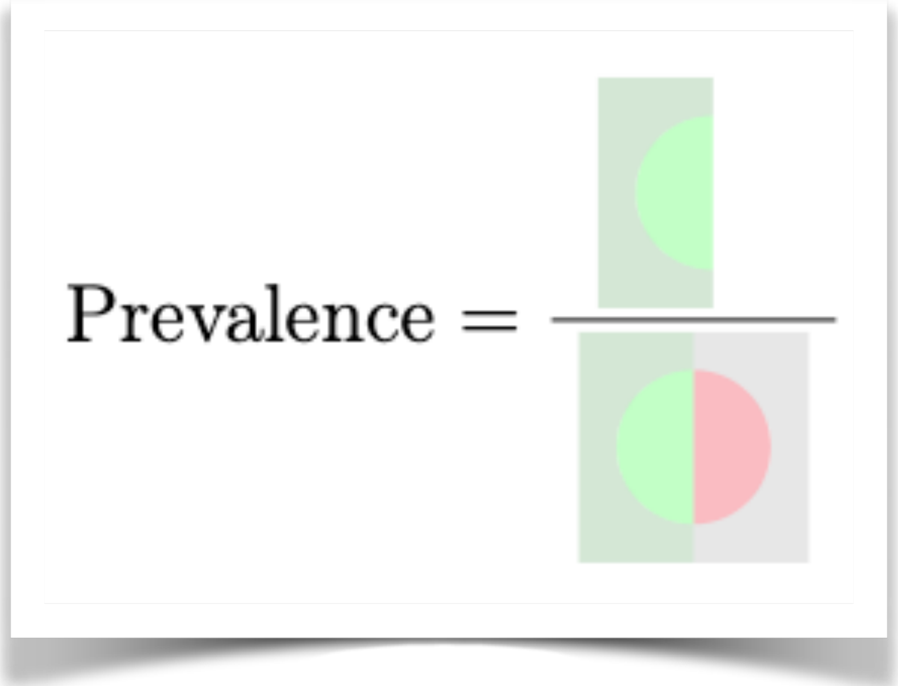
Predictive parity: same precision across groups



Unless ☝️

→ the prevalence is the same across groups OR

→ we have the perfect classifier



The Impossibility Theorem
mathematical formulation

$$FPR = \frac{p}{1-p}\frac{1-PPV}{PPV}(1-FNR)$$

if PPV is the same across groups (predictive parity/precision) but p (prevalence) differs between groups, we cannot achieve equal FPR and FNR across groups, equivalently

$$FPR = \frac{p}{1-p}\frac{1-PPV}{PPV}(TPR)$$

Reminder: $TPR = 1 - FNR$

Chouldechova, Big Data 2017

The Impossibility Theorem
mathematical formulation

$$FPR = \frac{p}{1-p}\frac{1-PPV}{PPV}(TPR)$$

Let's assume same PPV but different p between groups, so we can ignore PPV and rewrite the first term as x_i:

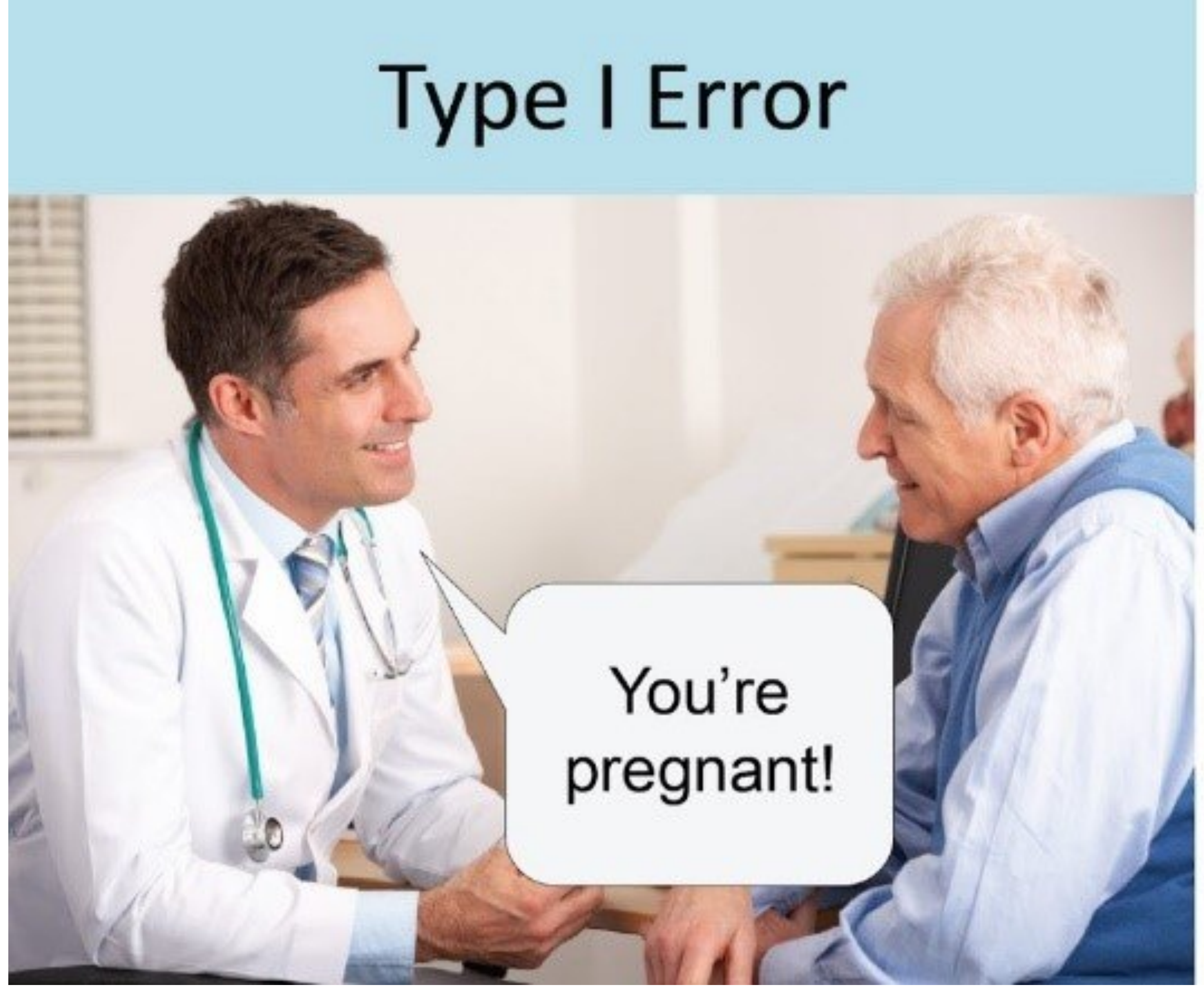
$$FPR_1 = x_1 \cdot TPR_1$$
$$FPR_2 = x_2 \cdot TPR_2$$

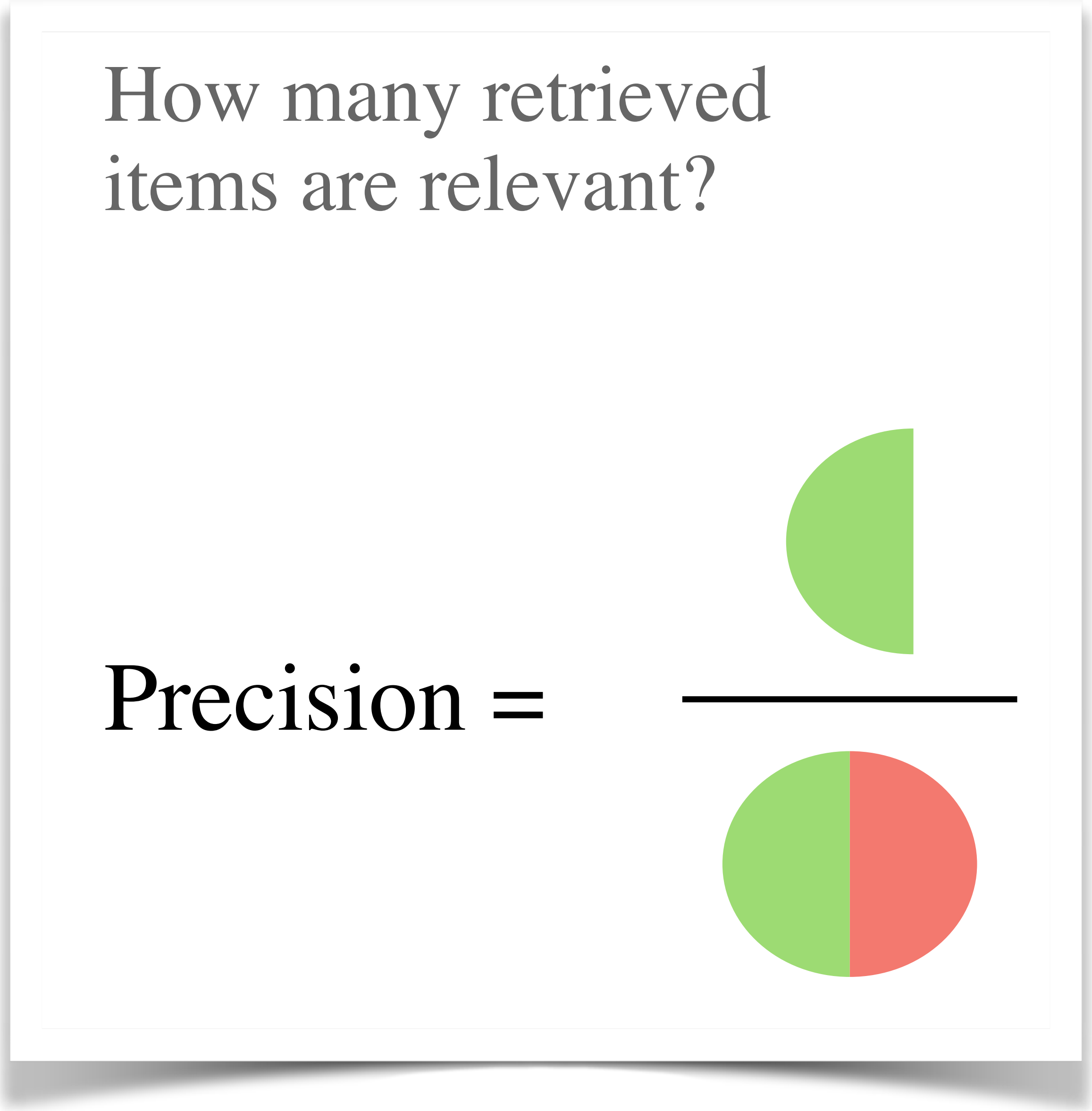
if $x_1 \neq x_2$, then $FPR_1, FPR_2, TPR_1, TPR_2$ cannot be the same.

Chouldechova, Big Data 2017

The Impossibility Theorem

alternative formulation



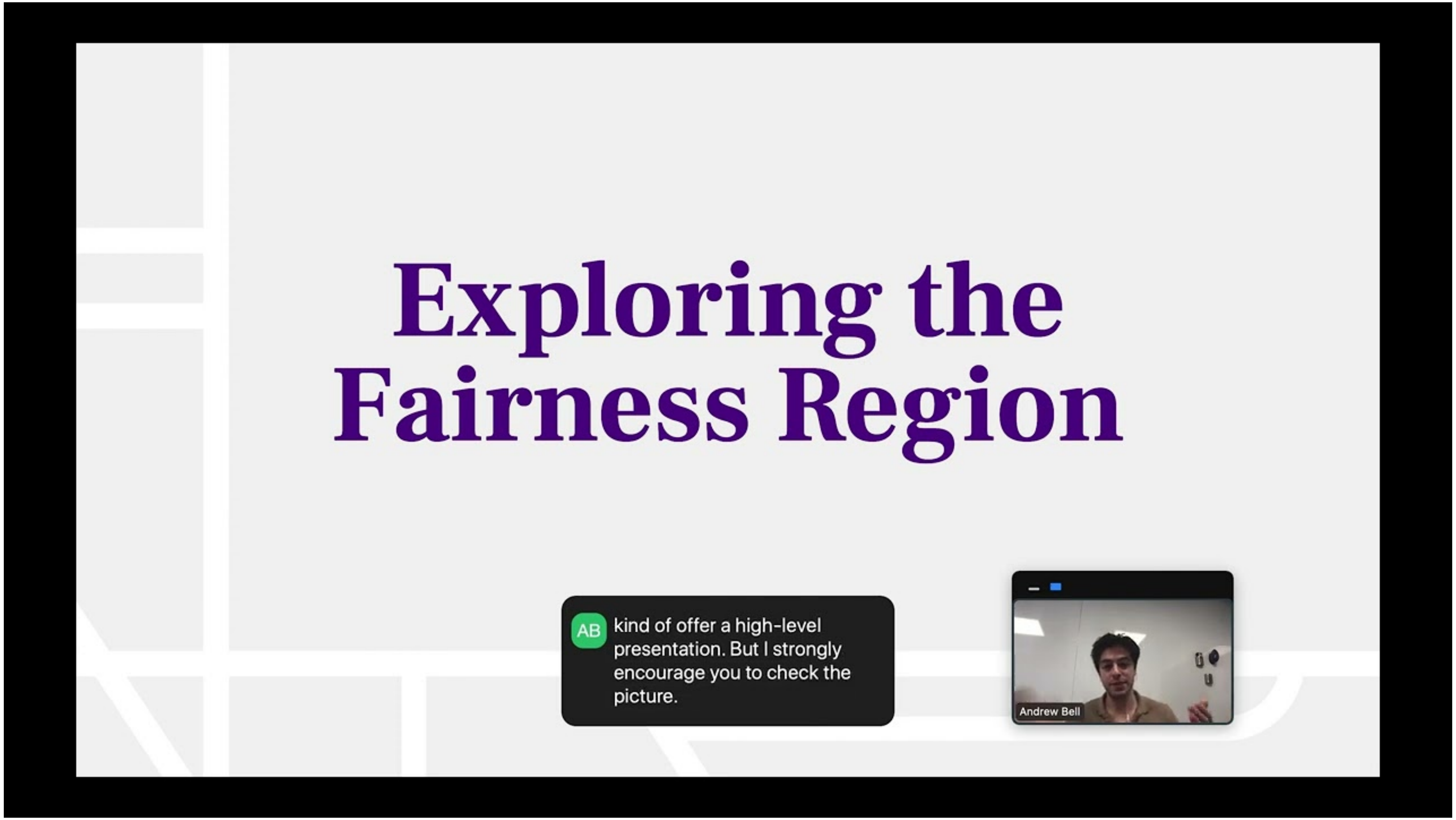


You cannot optimize for all at the same time!

What about including a margin of error?

Revisiting the Impossibility Theorem in Practice

The ~~Im~~Possibility Theorem



Bell et al, FAccT 2023

The Possibility Theorem

- although we cannot perfectly optimize for all 3 fairness constraints outside those two special cases (perfect classifier, equal prevalence) we might be able to find a solution that satisfies practitioners
- loosening constraints:

$$PPV_2 = PPV_1 + \epsilon_\vartheta$$
$$FPR_2 = FPR_1 + \epsilon_\alpha$$
$$FNR_2 = FNR_1 + \epsilon_\beta$$

- explore the space of feasible solutions

Bell et al, FAccT 2023

The Possibility Theorem

Datasets

Ukrainian External Independent Evaluation (EIE)

- standardized tests for secondary school graduates in Ukraine
- around 390k records
- sensitive attributes:
  student's sex and whether they live in a rural/urban area
- outcome:
  performance on multiple tests (ukrainian, math, geography, german, ...)

Bell et al, FAccT 2023

The Possibility Theorem
Datasets

Folktables

- individual-level and household-level data related to income, employment, health, transportation and housing in the US
- “millions” of records
- sensitive attributes:
  race and sex
- outcome:
  pre-defined prediction tasks (not further specified)

Bell et al, FAccT 2023

The Possibility Theorem

Results

shows for which values of k there is a set
of observations that satisfy fairness constraints
for FPR, FNR and PPV.

Table 1: Truncated experimental results.

Dataset (Outcome)	**Overall Prevalence (%)**	**Sensitive Attribute**	**Group Prevalence (%)**	**Maximum Prevalence Difference (%)**	**Optimal k Range (% Samples)**
EIE (Ukranian)	7.34	Sex	Female: 4.26; Male: 10.66	6.4	None
EIE (Ukranian)	7.34	Territory	Rural: 10.38; Urban: 6.3	4.08	[5,20]
EIE (Math)	31.1	Sex	Female: 30.92; Male: 31.26	0.34	All
EIE (Math)	31.1	Territory	Rural: 39.94; Urban: 28.12	11.82	[5,90]
EIE (Geography)	5.3	Sex	Female: 4.21; Male: 6.31	2.1	All
EIE (Geography)	5.3	Territory	Rural: 6.43; Urban: 4.87	1.56	All
EIE (German)	11.35	Sex	Female: 10.03; Male: 13.87	3.84	[5,90]
EIE (German)	**11.35**	**Territory**	Rural: 26.97; Urban: 8.57	**18.4**	**None**
Folktables (Employment)	46.45	Sex	Female: 44.3; Male: 48.74	4.44	All
Folktables (Employment)	46.45	Race	Asian alone: 50.0; Black or African American alone: 42.08; Other: 41.71; White: 47.36	8.29	All
Folktables (Travel Time)	53.78	Sex	Female: 51.72; Male: 55.8	4.08	All
Folktables (Travel Time)	**53.78**	**Race**	Asian alone: 66.86; Black or African American alone: 69.15; Other: 64.63; White: 48.51	**20.64**	**[5,55]**

Bell et al, FAccT 2023

The Possibility Theorem

Results

larger k ranges for prevalences
closer to 50% and for
smaller group prevalence differences

shows for which values of k there is a set
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Bell et al, FAccT 2023

The Possibility Theorem
Summary

- mathematically, we cannot achieve equalized odds and predictive parity at the same time
  - except for equal prevalence or the perfect classifier
- if we allow for a margin of error, it is possible to optimize for both
- Bell et al. showed that the space of possible solutions is bigger if
  - prevalences are close to 50%
  - the difference in prevalences is small

Overall summary

- there is not THE definition of fairness
- there is also a more in-depth philosophical debate that we didn't touch upon
- group fairness, equalized odds and predictive parity are the most common definitions
- depends on the context which one to use or optimize for
- often we are trying to optimize for multiple at a time